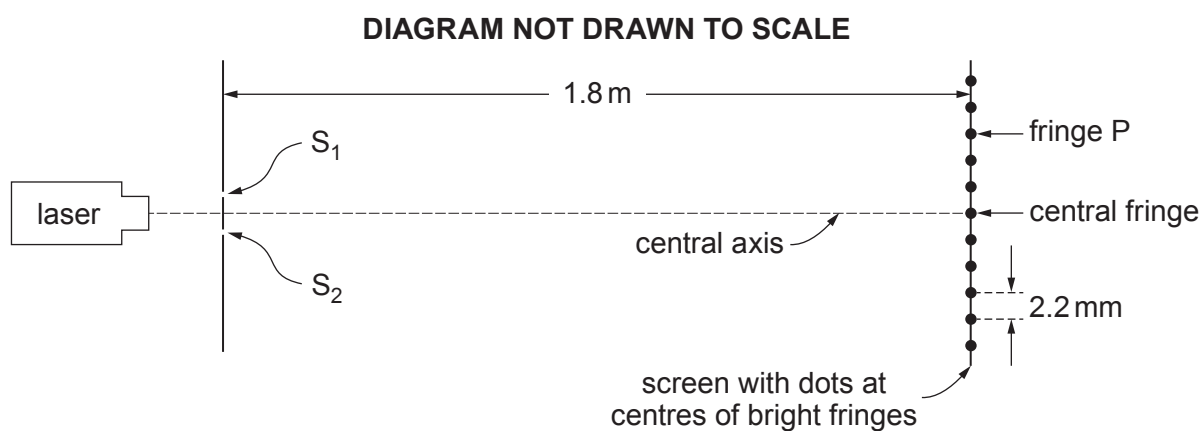


5. (a) In the two-slit set-up shown, the centres of the bright fringes are 2.2 mm apart. S_1 and S_2 are slits with centres 0.45 mm apart, acting as in-phase sources.



- (i) Explain the role of diffraction in producing the fringe pattern. [2]

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- (ii) Calculate a value for the wavelength of the laser light. [2]

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- (iii) Using your answer to (a)(ii) determine the path difference $S_2P - S_1P$ (see diagram), giving your reasoning. [3]

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- (b) Give **two** reasons why a diffraction grating would be expected to give a more precise value for the wavelength than the two-slit method. [2]

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